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Post-Quantum Signature Experiments and Migration Considerations for the Resource Public Key Infrastructure (RPKI)

Abstract

This document reports experiments with post-quantum signature algorithms and analyzes migration approaches for the Resource Public Key Infrastructure (RPKI). The experiments compare classical, post-quantum, and composite signature candidates; generate and validate RPKI-profiled certificate, CRL, manifest, and ROA test objects; compare separately produced Current Suite and Next Suite validation outputs; and demonstrate an experimental Mixed Tree. The evaluation also models an idealized parallel-publication footprint and estimates RRDP delta and snapshot response-body sizes from captured RPKI data. The results identify implementation, interoperability, repository-distribution, and operational questions that need to be resolved before a production algorithm profile or transition procedure can be specified. This document is informational. It does not update RFC 7935 or RFC 6916, define a new RPKI algorithm profile, or authorize the use of the evaluated algorithms or Mixed Tree migration in the production RPKI.

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1. Introduction

The RPKI relies on digital signatures in resource certificates, CRLs, certification requests, and CMS signed objects such as manifests and Route Origin Authorizations (ROAs). The deployed RPKI algorithm profile is based on RSA with SHA-256. A Cryptographically Relevant Quantum Computer (CRQC) capable of executing the required quantum algorithms would undermine the unforgeability of the RSA signatures used by the deployed RPKI profile. Preparing for that risk requires measurements and interoperable implementations before a production transition procedure can be selected.

This document records that preparatory work. The evaluation also examines the impact of post-quantum signatures on repository size and distribution using observed RPKI data. Certification Authority (CA) and Relying Party (RP) implementations require support for the additional signature algorithms. The Mixed Tree experiment also evaluates a certificate-path construction that is not permitted by the migration procedure in RFC 6916. Routers that consume Validated ROA Payloads (VRPs) through the RPKI-Router Protocol (RTR) [RFC8210] or local files do not process the evaluated signatures directly.

The descriptions of algorithms, encodings, validation behavior, and migration steps in this document are experimental assumptions and observations. They are not interoperability requirements for production RPKI implementations.

This document does not select or recommend a production PQ algorithm. The candidate algorithms and experimental results explore the problem space and identify requirements and trade-offs for a future RPKI PQ migration. The eventual production algorithm choice should be made separately when sufficient cryptographic, implementation, and operational information is available. Investigation of PQ migration can continue while this choice remains open.

2. Terminology

This document uses the terminology of the RPKI architecture [RFC6480], the resource certificate profile [RFC6487], the RPKI signed object template [RFC6488], the RPKI algorithm agility procedure [RFC6916], and the RPKI algorithm profile [RFC7935].

Current Suite: The algorithm suite specified by the currently applicable production RPKI algorithm profile. At the time of writing, this is RSA-2048/SHA-256 as profiled by RFC 7935.

Next Suite: A candidate algorithm suite that is implemented and tested before it becomes the Current Suite.

Mixed Tree: An experimental RPKI certification hierarchy in which a parent CA using the Current Suite signs a child CA certificate whose subject public key uses the Next Suite, allowing the child subtree to use the Next Suite while its ancestors remain on the Current Suite.

Parallel Publication: A migration approach in which products that correspond as defined in [RFC6916] are published under both the Current Suite and the Next Suite during a transition period.

Composite Signature: A signature construction that combines multiple component algorithms into one algorithm identifier and signature value. In the LAMPS construction evaluated by this document, verification succeeds only when every component signature validates.

3. Scope

The algorithm-profile analysis in this document applies to RPKI resource certificates, CRLs, certification requests, BGPsec Router Certificates [RFC8209], and the CMS signed objects that reuse the RPKI signed object template [RFC6488], including manifests [RFC9286], ROAs [RFC9582], Signed Checklists (RSC) [RFC9323], ASPA objects [I-D.ietf-sidrops-aspa-profile], and Trust Anchor Key (TAK) objects [RFC9691]. The generated signed objects in these experiments cover manifests and ROAs; additional object types are listed in the Implementation Status section. The CMS signed objects are treated as a single signed-object algorithm profile, as described in the following section.

This document covers the RPKI signatures on BGPsec Router Certificates, but does not define or change the BGPsec UPDATE signature algorithm specified by [RFC8608]. For a BGPsec Router Certificate, the experiment applies only to the certificate signature made by the issuing RPKI CA. The subject public key algorithm remains governed by the BGPsec UPDATE algorithm profile and is not required to match the RPKI certificate-signature suite.

This document does not specify changes to RTR, TAL formats, RRDP [RFC8182], rsync, the RPKI Certificate Policy, or the BPKI used to authenticate provisioning and publication relationships established through [RFC8183]. BPKI algorithm migration is a related operational problem, but is not evaluated by this document and need not use the same algorithm or transition schedule as RPKI repository objects. The repository-distribution evaluation estimates RRDP delta and snapshot response-body sizes; equivalent rsync and Erik Synchronization comparisons remain future work.

4. Signed Object Coverage

Manifests [RFC9286], ROAs [RFC9582], Signed Checklists [RFC9323], ASPA objects [I-D.ietf-sidrops-aspa-profile], TAK objects [RFC9691], and any future object types built on the RFC 6488 template share one CMS structure, one EE certificate model, one certification infrastructure, and one repository. The evaluation model therefore treats them as one signed-object algorithm profile. Migrating, for example, ROAs to a PQC suite while leaving ASPA objects on RSA would introduce per-object-type algorithm diversity, leaving part of the RPKI signed-object set dependent on the Current Suite.

Changing only the CMS signature algorithm does not change an object's eContentType or object-specific payload syntax. Some objects, notably TAK objects, may nevertheless carry public keys whose algorithms change as part of a wider trust-anchor migration and therefore require

separate object-specific interoperability testing. A future standards-track profile that selects a new mandatory RPKI algorithm suite would update [RFC7935]. An object-specific RFC needs an update only if that object's payload or validation semantics also change; this document makes no such change.

BGPsec UPDATE signatures are not RFC 6488 signed objects and are outside this experiment. BGPsec Router Certificates and their covering CRLs and manifests are repository products and remain part of the evaluation scope.

5. Relationship to RFC 6916

RFC 6916 specifies a top-down algorithm transition using parallel certification hierarchies. Section 4.2 of [RFC6916] states that a CA "MUST NOT" sign a CA certificate whose subject key corresponds to an algorithm suite different from the suite used to sign that certificate.

The Mixed Tree experiment described in this document deliberately evaluates the X.509 construction excluded by RFC 6916. Objects produced by this experiment are therefore confined to test TALs and test repositories. Production use of this construction would require separate Standards Track work updating or replacing the applicable algorithm profile and migration procedure.

6. Evaluation Goals

The experiments use the following evaluation goals.

- Evaluate post-quantum and composite signatures within the existing RPKI object and validation model, reusing LAMPS PKIX and CMS encodings without introducing new object formats.
- Keep routers as consumers of validated payloads and apply a common signature suite to all RFC 6488 signed objects.
- Compare Parallel Publication and Mixed Tree migration, including compatibility and the operational effects of algorithm coexistence.
- Measure cryptographic performance, repository size, and distribution impact, and compare validated outputs across algorithm suites.
- Document reproducible evidence, implementation limitations, and remaining interoperability and operational questions.

7. Experimental Migration Approaches

This section describes the two migration approaches evaluated in this document. The signature suite can be chosen independently of the migration approach.

An RPKI cryptographic transition incurs implementation, testing, and operational costs that may be distributed across RP and CA software implementers, RIRs and other CA operators, network operators, vendors, and hardware and HSM providers. These costs, who bears them, and the

effort needed to coordinate deployment and maintain service during transition are relevant when choosing an algorithm and migration strategy. This document does not resolve how those costs should be funded.

7.1. Parallel Publication

RFC 6916 specifies a top-down transition in which a parent CA adopts support for the Next Suite before its children. During phases 2 and 3, corresponding Current Suite and Next Suite product sets are maintained in parallel. RFC 6916 deliberately avoids mixed-suite CA certificates: a CA certificate signed using one suite does not carry a subject key associated with another suite.

The Parallel Publication experiment compares validation outputs from separately produced Current Suite and Next Suite repositories and models an idealized duplicate footprint. It does not implement concurrent maintenance of corresponding products through the complete RFC 6916 phase-by-phase operational transition. It does not define new payload encodings for manifests, ROAs, or CRLs. As described in Section 6 of [RFC6916], RPs can retrieve product sets under both algorithm suites and compare their outputs for testing. This experiment performs that comparison using the resulting VRP sets and CCR [I-D.ietf-sidrops-rpki-ccr].

The experimental harness needs a mapping between corresponding products for measurement and debugging. Such a mapping can be derived from the publication point structure, object names, CA hierarchy, or an implementation-specific record. It is not proposed as a new on-wire RPKI object.

7.2. Mixed Tree

A parent using the Current Suite signs a child CA certificate whose subject public key belongs to the Next Suite. The child then issues its certificates, CRLs, and CMS signed objects using the Next Suite without first requiring the parent to migrate its own CA key and product set. Production issuance would still require the parent to process the child's Next Suite certificate request and proof of possession. The fixture generator issues the child certificate from a locally generated PKCS#10 request using OpenSSL. The Krill experiment creates and verifies a Composite CSR through a same-server issuance path that bypasses the RFC 6492 CMS/HTTP exchange; independent provisioning interoperability is not established [RFC6492].

The Mixed Tree migration approach is independent of the selected signature suite. The evaluated repository instantiates the Next Suite with the Composite ML-DSA configuration evaluated in this document. This construction is not permitted by the RFC 6916 transition procedure and is evaluated only as an experimental alternative under test TALs.

Unlike the RFC 6916 procedure, a switched subtree does not maintain corresponding Current Suite and Next Suite products. This avoids duplicate repository content and the publication, configuration, and rollover work needed to keep two product sets aligned. It also permits subtrees to move independently after the necessary parent and RP support is available, rather than waiting for every ancestor to migrate its own CA key and product set.

This introduces a compatibility trade-off: once a subtree switches, an RP that does not support the Next Suite cannot validate it through the certification path. Mixed Tree deployment therefore replaces the parallel legacy hierarchy with a requirement for sufficient RP support before each subtree is migrated. The complete certification path remains dependent on Current Suite certificate signatures above the migration boundary.

Migration model	Migration ordering	Legacy RP compatibility	Parallel products	Principal limitation
RFC 6916 Parallel Publication	Top-down	Current Suite hierarchy remains available	Required during transition	Repository and operational duplication
Mixed Tree	Per subtree after required support is available	Unsupported RPs lose the switched subtree	Not required within the switched subtree	The path still depends on Current Suite ancestors

Table 1

The experiment uses test repositories and test TALs. It does not define or authorize a production transition procedure.

8. Evaluated Algorithm Suites

8.1. Current Suite

The Current Suite baseline used by this experiment is RSA PKCS #1 v1.5 with SHA-256 as specified by RFC 7935. This document does not set a production transition timetable.

8.2. Evaluated Composite Configuration

The composite signature construction used by this experiment is defined in revision 19 of [I-D.ietf-lamps-pq-composite-sigs]. The evaluated configuration is id-MLDSA65-ECDSA-P256-SHA512 (OID 1.3.6.1.5.5.7.6.45), whose identifier and component-algorithm combination are specified by that document. It combines ML-DSA-65 with ECDSA P-256 and requires both component signatures to validate. The corresponding CMS use is specified by [I-D.ietf-lamps-cms-composite-sigs]. The choice of this composite configuration is an experimental input, not a recommendation for production deployment.

The object experiment applies [I-D.ietf-lamps-pq-composite-sigs] to composite public keys and certificate signatures, and [I-D.ietf-lamps-cms-composite-sigs] to composite CMS SignedData. The public reference implementation has generated complete composite X.509 certificates, CRLs, ROAs, and manifests and has validated a Mixed Tree with an RSA parent and a Composite child using experimental rpki-client and Routinator extensions. The two RP processing paths share one OpenSSL and Composite provider backend, so this is not independent cryptographic interoperability evidence.

Experiments with the composite suite are confined to isolated repositories under test TALs. Experimental objects, production repositories, production keys, and production TALs are deliberately kept separate.

8.3. Additional Candidate Suites

ML-DSA-44 is an additional experimental configuration. It produces smaller public keys and signatures and may sign and verify faster, which matters in a system where RPs repeatedly synchronize repository state and validate large numbers of objects. The experiment uses ML-DSA-65 in the composite configuration as an explicit experimental choice, not because the RPKI requires NIST security Category 3. The Rationale for the Experimental Configurations section discusses this choice and its counterarguments.

ML-DSA-87 is included as a higher-security comparison candidate. It is more conservative than ML-DSA-65 but carries correspondingly larger size and performance costs.

SLH-DSA-SHAKE-128s and SLH-DSA-SHAKE-192s are included for cryptographic-diversity comparison. Their signature algorithm, PKIX, and CMS specifications are defined by [FIPS205], [RFC9909], and [RFC9814]. They are not proposed as the initial experimental suite because their signature sizes are substantially larger than those of ML-DSA in the evaluated configurations. Their primitive and certificate/CRL measurements do not include end-to-end RP validation or global repository/RRDP projections.

Supplementary primitive benchmarks include experimental Falcon, MAYO, and SNOVA, and certificate/CRL measurements include experimental Falcon-512. These configurations remain unevaluated for end-to-end RP validation and the global repository/RRDP projections. The Rationale for the Experimental Configurations section discusses FN-DSA further because its compact signatures make it an attractive candidate for the RPKI's bulk validation model.

Other Composite ML-DSA combinations specified by LAMPS remain candidates for comparison. Changing the component pair would change object sizes and cryptographic costs, but would not determine whether a deployment uses Parallel Publication or Mixed Tree migration. RP support for a selected suite remains a deployment prerequisite.

8.4. Classical Reference Points

To put the cost of PQC candidates in context, this document uses two compact classical algorithms as non-normative reference points: ECDSA P-256 with SHA-256 [FIPS186-5], which is already used for BGPsec UPDATE signatures [RFC8608], and Ed25519 [RFC8032]. Neither is a CRQC-resistant algorithm, and neither is proposed here as an RPKI suite. They provide compact classical reference points for signature and key sizes in the classical setting: RSA-2048 public keys and signatures are several times larger than those of these curves, and PQC candidates should be compared against both baselines rather than against RSA alone.

8.5. Evaluation Dimensions

The evaluation considers the following dimensions: certificate and CRL size under the RPKI profile; CMS signed object size; signing and verification cost under the RP workload; impact on repository size and RRDP response-body sizes; CA key rollover and publication cycle impact; HSM support; and standardization and implementation maturity.

RP workloads are dominated by verification: an RP verifies repository objects but signs nothing as part of validation. The deployed RSA profile is particularly inexpensive to verify, so increases in verification cost are especially relevant to RP operation. The source-calibrated projections in Appendix A put ML-DSA-44 at 3.52 times and ML-DSA-65 at 4.65 times the observed RSA DER footprint. These projections do not cover every candidate algorithm.

Appendix A gives the measurements and open measurement tasks. Measured values depend on the implementation and environment; they are not protocol requirements.

8.6. Rationale for the Experimental Configurations

This experiment evaluates a composite suite because a global RPKI migration may benefit from not depending exclusively on a newly deployed PQC algorithm. The LAMPS construction accepts a composite signature only when both component signatures validate. Under the assumptions stated by that construction, its EUF-CMA guarantee is retained when at least one component remains EUF-CMA secure and the prehash remains collision resistant.

A CRQC defeats the ECDSA component, so security against a quantum adversary still depends on ML-DSA-65 remaining secure. The composite suite also does not protect against failures shared by both components or by the combiner, encoding, key management, or validation implementation. It does not in general preserve strong unforgeability when only one component has that property. An implementation defect in one component can be tolerated only when the other component, the combiner, encodings, key handling, and the verification of all component signatures remain unaffected.

ML-DSA-65 is used as the PQC component of the evaluated composite configuration because it has a final FIPS signature specification [FIPS204], corresponding PKIX [RFC9881] and CMS [RFC9882] algorithm identifier specifications, and implementations available in the software environment used by this experiment. The experiment uses Category 3 as an experimental choice to examine the cost of a larger security margin in a system where another global algorithm migration could take multiple years. This is not an RPKI requirement. It was not selected for minimum size or maximum speed; it is neither the smallest nor the fastest possible signature algorithm.

The choice between ML-DSA-44 and ML-DSA-65 remains open. The argument for ML-DSA-65 is conservatism: the RPKI is a single global system, algorithm migrations in it are slow and expensive, and a larger security margin reduces the probability of needing another migration. The argument for ML-DSA-44 is that a structural cryptanalytic break of module lattices would likely affect all ML-DSA parameter sets. On this view, the extra category mainly protects against

gradual erosion of concrete security estimates rather than a qualitative break. The roughly 25-35% smaller keys and signatures of ML-DSA-44, or a composite configuration using ML-DSA-44, may therefore be a better use of the size budget [Doesburg2025]. The experiments use ML-DSA-65 in the measured composite configuration, keep ML-DSA-44 in the comparison, and leave the parameter-set choice as a question for further work.

ML-DSA-87 provides a higher-security comparison point, but its larger signatures and higher verification cost warrant further evaluation before deployment.

The experimental Falcon-512 encoding used for FN-DSA-512 comparison has signatures roughly one fifth the size of ML-DSA-65 signatures; this is a size comparison, not a comparison at equal security categories. Both published RPKI analysis [Doesburg2025] and repository-scale redesign work [pqRPKI] identify Falcon as the compact lattice option. FN-DSA composite configurations, end-to-end RP validation, and global repository/RRDP projections remain future work for the following reasons:

- FIPS 206 remains under development [NIST-FN-DSA-Plan], and the LAMPS PKIX and CMS profiles remain works in progress [I-D.ietf-lamps-fn-dsa-certificates] [I-D.ietf-lamps-cms-fn-dsa]. The measured Falcon-512 encoding is not a final FN-DSA profile.
- Side-channel-resistant FN-DSA implementations and HSM support require separate implementation and deployment evaluation.
- Availability of FN-DSA in the platforms that RPKI CAs, RIRs, HSM vendors, and validator implementations actually use is a separate question from the algorithm's intrinsic merits. This experiment does not evaluate production RPKI CA or RP support for FN-DSA.

These are reasons to sequence the evaluation, not to dismiss the algorithm. FN-DSA remains in the comparison set as the compact signature candidate. Conditions for further evaluation are listed in the Questions for Further Work section.

Algorithm selection for the RPKI cannot be based on software benchmarks alone. HSM support for a candidate algorithm is also a deployment prerequisite for CAs that protect their signing keys in HSMs.

8.7. Risk of Premature Algorithm Selection

Algorithm selection itself carries a deployment risk. An RPKI algorithm transition requires substantial implementation and operational effort across the ecosystem. If the selected algorithm later proves unsuitable due to security or operational issues, another ecosystem-wide transition may be required. Even if it remains secure, the emergence of a significantly more efficient algorithm could make the original choice costly in hindsight.

The set of practical PQ algorithms and their properties is still evolving. A future algorithm may offer materially better size, performance, implementation, hardware, or operational characteristics. The size, performance, and repository results in this document illustrate relevant trade-offs but do not establish which algorithm should be selected for production. Algorithm

maturity and deployment experience, as well as cryptographic security, should therefore inform the choice and its timing. Candidate evaluation should explicitly consider the cost of an unsuitable selection and a subsequent ecosystem-wide transition.

9. Experimental Object Encoding

9.1. Certificate and CRL Encoding

The composite certificate and CRL experiment follows the encodings in [I-D.ietf-lamps-pq-composite-sigs]. Under that assumption, a composite subject carries id-MLDSA65-ECDSA-P256-SHA512 in the SPKI AlgorithmIdentifier with absent parameters. A certificate or CRL signed by the composite issuer uses that identifier in its signatureAlgorithm field. A child CA certificate signed by a Current Suite issuer instead retains the issuer's Current Suite signatureAlgorithm while carrying the composite identifier in the subject SPKI.

The same composite SPKI and signature encodings are assumed for resource certificate requests and their proof-of-possession signatures. Pure ML-DSA certificates and CRLs used for component measurements follow [RFC9881]; they provide component-level evidence and are not proposed as a production Next Suite. Local OpenSSL issuance and the Krill same-server path exercise Composite request signing and verification, but not an independent RFC 6492 CMS/HTTP provisioning exchange.

The generated RPKI test objects preserve the existing keyUsage constraints. CA certificates carry keyCertSign and cRLSign, while EE certificates for signed objects carry digitalSignature and are not used as CA certificates. The experiment does not change resource extension semantics, the certificate policy OID, certificate path validation, manifest processing, or CRL processing.

9.2. CMS Signed Object Encoding

The composite CMS experiment combines [I-D.ietf-lamps-cms-composite-sigs] with the RPKI signed object template in [RFC6488], as updated by [RFC9589]. The assumed SignerInfo signatureAlgorithm is id-MLDSA65-ECDSA-P256-SHA512 with absent parameters. The SignedData digestAlgorithms set contains exactly one AlgorithmIdentifier, id-sha512, and the SignerInfo digestAlgorithm is id-sha512, retaining the single-digest cardinality in Section 2.1.2 of [RFC6488]. The parameters are absent in both AlgorithmIdentifiers, and the message-digest signed attribute contains the SHA-512 digest of the eContent. Using SHA-512 here is an experimental departure from the deployed SHA-256 profile in [RFC7935], not authorization for production use.

The experiment retains the signedAttrs restrictions from the existing RPKI signed object profile: one content-type attribute, one message-digest attribute, and one signing-time attribute, with no binary-signing-time or other signed attributes. The eContentType, eContent, and object-specific validation rules remain unchanged.

The implementation generated and validated complete Composite ROA and manifest SignedData using these encodings, together with the certificates and CRLs required for repository validation. Experimental rpki-client and Routinator extensions processed the resulting repository. Interoperability with a cryptographic implementation independent of the shared OpenSSL and Composite provider remains open work.

9.3. Manifest Processing During Migration

A manifest covers the products of one CA instance at one publication point, as specified by [RFC9286] and updated by [RFC9981]. The manifest is signed with a one-time-use EE certificate issued by that CA. Its fileList contains the certificates issued and published by that CA, the CA's current CRL, and signed objects whose embedded EE certificates were issued by that CA.

The relevant RP check is therefore issuer and publication-scope consistency, not equality between the manifest signing key and product keys. An RP validates the manifest EE certificate under the associated CA, verifies each listed certificate, CRL, or signed object under that same CA instance as required by its object profile, and checks the publication point, file name, and file hash according to [RFC9286]. A shared publication point can contain products from multiple CA instances during key rollover, but each manifest covers only its associated CA instance.

Mixed Tree migration and composite signatures do not change these checks. This document therefore introduces no additional requirement for the manifest EE key to equal a key used by a listed product, and it does not weaken the existing RP checks that bind every listed product to the manifest's CA scope.

10. Related Experimental Designs

The cryptographic-object and Mixed Tree experiments preserve the existing X.509 resource-certificate and CMS signed-object model while changing signature algorithms and, in the Mixed Tree case, the certification path. The Null Scheme [I-D.doesburg-sidrops-nullscheme] preserves the signed-object structure but replaces the one-time-use EE key pair and CMS signature with a public key derived from the message digest and an empty signature, reducing redundant cryptographic cost.

pqRPKI [pqRPKI] instead introduces a Merkle Tree Ladder and restructures manifest, delegation, and bulk-verification processing. It is an alternative repository architecture rather than a direct algorithm substitution. Comparing these object-design alternatives requires separate measurements; the cryptographic-object experiments in this document cover only the structure-preserving approach.

11. Repository Distribution Considerations

11.1. Impact of Larger Objects

Changing the signature suite affects both cryptographic processing and repository distribution. Larger certificates, CRLs, manifests, and signed objects increase the bytes transferred during initial repository synchronization and in updates that contain those objects. For the observed cache assessed in Appendix A, full replacement with the evaluated Composite suite is projected to increase the DER footprint from 1.017 GiB to 4.825 GiB. Transport scalability is therefore a deployment constraint that needs to be evaluated separately from algorithm correctness and RP validation time.

11.2. RRDP, rsync, and Erik Synchronization

RRDP [RFC8182] distributes snapshots and a journal of deltas. An RP obtains a complete snapshot when the notification file no longer offers a contiguous delta chain from the RP's serial, or when the session identifier changes. A polling RP can receive intermediate publication states that it does not ultimately use. An rsync synchronization first exchanges repository metadata and then transfers changed files; even an unchanged repository therefore has a cost that grows with the file list.

Erik Synchronization [I-D.ietf-sidrops-rpki-erik-protocol] is an RPKI-specific replication protocol using Merkle trees, content-addressable naming, and HTTP. An RP compares ErikIndex and ErikPartition objects and fetches selected objects by hash. Snapshot and segment prefetching can reduce the request cost of cache bootstrap and catch-up. This document does not specify Erik or require it for PQC deployment; equivalent trace-backed rsync and Erik comparisons remain separate work.

11.3. Transport Experiment Boundary

Appendix A projects complete DER object sizes from a public RPKI cache and evaluates captured RRDP delta and snapshot XML bodies for RSA-2048, ML-DSA-44, ML-DSA-65, and the evaluated Composite suite. The delta and snapshot samples have different capture dates and are reported separately. These results quantify object and response-body sizes, not production-network throughput or per-client traffic.

12. Experimental Results

12.1. Results Summary

OpenSSL 3.6.2 and the evaluated Composite provider generated complete certificate, CRL, manifest, and ROA sets for pure ML-DSA-65, Composite ML-DSA, and a Mixed Tree with an RSA parent and a Composite child. Experimental rpki-client and Routinator extensions validated each repository and produced the same two VRPs as the RSA baseline. An experimental Krill extension then created a Composite child below an RSA parent, published and replaced ROAs, and rolled

the child back to RSA. Both experimental RPs derived the expected VRPs at each stage. The Krill test used same-server issuance with CSR signature verification, not a complete external RFC 6492 provisioning exchange.

Appendix A reports the cryptographic benchmarks, complete-DER repository projections, and captured RRDP body-size projections, with their methods and limitations.

The two RP extensions use the same OpenSSL and Composite provider backend. The experiment therefore demonstrates two RP processing paths, but not independent cryptographic interoperability or production readiness.

12.2. Relying Party Evaluation

The experiment extended rpki-client and Routinator to recognize the id-ml-dsa-65 and id-MLDSA65-ECDSA-P256-SHA512 identifiers and to delegate their cryptographic operations to OpenSSL providers. Each RP processed four complete repositories: the RSA baseline, pure ML-DSA-65, Composite ML-DSA, and a Mixed Tree with an RSA parent and a Composite child. For every repository, the test covered certificate-path and CRL validation, manifest and CMS validation, ROA processing, and VRP production.

The Mixed Tree test additionally verified that the issuer signature algorithm and subject SPKI algorithm are processed independently. The RSA parent's public key verifies the signature on the child CA certificate, while the Next Suite public key carried in that certificate's SPKI is used to validate signatures issued by the child.

Both experimental RPs also produced the expected VRPs for controlled repositories with an RSA parent and a Composite child publishing up to 1,000 ROAs. A separate synthetic topology used one RSA parent and 100 Composite child CAs, each publishing one ROA. The repository contained 403 objects and produced 100 VRPs with both experimental RPs. Removing one child publication point reduced the result to 99 VRPs without affecting the remaining children.

When an experiment validates both Current Suite and Next Suite products, it separately compares their validated outputs. For ROAs, the comparison uses equality of the canonical VRP sets by prefix, maxLength, and origin AS. When both runs use the same CCR version and hash algorithm, equal ROAPayloadState hashes indicate equality of those canonical VRP sets under the collision-resistance assumption of the CCR digest [I-D.ietf-sidrops-rpki-ccr]. The experiment parsed actual rpki-client CCR DER for the RSA, pure ML-DSA-65, Composite, and Mixed Tree repositories and recomputed every embedded collection hash. All four ROAPayloadState hashes were equal, while ManifestState differed as expected. TrustAnchorState differed because the four test runs used distinct TA public keys and SKIs. This result uses one RP implementation that produced CCR output. ROAPayloadState does not preserve per-VRP certificate-chain or publication provenance, so comparing the certificate chain and publication source associated with each VRP requires additional experiment records. Differences in validated outputs are recorded as an experimental result rather than silently merged.

This document does not require routers to support PQC. Routers receive validated payloads through RTR or local export formats; the algorithm migration is intended to leave the semantic content of those outputs unchanged.

13. Implementation Status

This section records the status of the experiments at the time of posting this Internet-Draft, using the reporting pattern described in [RFC7942]. It is intended to assist IETF discussion and is to be removed before publication as an RFC.

pqc-rpki-lab is a research prototype used for the experiments described in this document.

Implemented:

- Generation of complete pure ML-DSA-65 and Composite certificates, CRLs, manifests, and ROAs, including a Mixed Tree repository with an RSA parent and a Composite child.
- Experimental rpki-client and Routinator extensions that validate the RSA, pure ML-DSA-65, Composite, and Mixed Tree repositories and produce the expected VRPs.
- Experimental Krill same-server issuance with Composite CSR/PoP processing, publication, one-ROA replacement, and RSA rollback for a Composite child below an RSA parent.
- Complete-DER repository-size projections and captured RRDP delta and snapshot response-body size accounting described in Appendix A.
- Fifteen negative cases for cryptographic and profile failures, seven cases for repository-operation failures, tests showing that removal of one child publication point leaves sibling VRPs unaffected, and CCR output comparisons.

Not yet implemented or incomplete:

- Independent RFC 6492 provisioning interoperability with Composite certificate requests and proof of possession, BGPsec Router Certificates, and Composite ASPA, RSC, and TAK objects.
- Independent cryptographic interoperability and a second RP implementation capable of producing CCR output.
- Re-signing of a multi-CA corpus modeled on the public RPKI and production-network RRDP, rsync, and Erik measurements.
- HSM support, long-running RP resource use, and additional candidate suites.

The highest-priority gaps are independent cryptographic interoperability and validation over a multi-CA corpus modeled on the public RPKI that can be re-signed under each evaluated configuration.

13.1. Upstream Library Support Snapshot

This snapshot of upstream library support was updated on 25 October 2026. Released library functionality is distinguished from the lab extensions above; it does not establish production RPKI CA or RP support, RPKI profile conformance, or independent interoperability. Support for a PQ signature primitive alone does not establish X.509 or CMS SignedData support, and pure PQ signatures are distinct from composite or hybrid constructions.

- OpenSSL: The released 3.5 series introduced pure ML-DSA and SLH-DSA primitives [OpenSSL-3.5]. Upstream release tests cover ML-DSA X.509 certificate generation [OpenSSL-X509-Tests] and CMS signing and verification using X.509 signer certificates with ML-DSA and SLH-DSA [OpenSSL-CMS-Tests]. SLH-DSA certificate-signature coverage has not been separately verified for this snapshot. These pure PQ capabilities are separate from the external Composite provider used by this experiment; the lab results do not establish native production support for the evaluated Composite profile.
- LibreSSL: Version 4.4.0 was released as a stable release on 22 October 2026 [LibreSSL-Releases]. Its release notes describe hybrid TLS key establishment using X25519MLKEM768, which does not establish PQ signature support. The reviewed 4.4.0 release notes do not establish ML-DSA or SLH-DSA signature support, either for X.509 or CMS SignedData. Such support remains unverified in this snapshot; this is not a claim that no implementation exists.
- Bouncy Castle: The released Java 1.86 source includes tests that generate pure ML-DSA and SLH-DSA X.509 certificates and create and verify CMS SignedData [BC-PQC-Tests]. Its release notes also record composite signature support, including CMS support added in 1.83 [BC-Java-Release-Notes]. Released composite functionality should not be confused with a final algorithm profile: compatibility with the specific LAMPS revisions and Composite configuration evaluated here remains to be verified. This snapshot concerns the regular Java release, not every Bouncy Castle language or FIPS product.

14. Security Considerations

This document addresses forgery of RPKI signatures in the presence of a CRQC. Existing considerations for CA compromise, repository compromise, operational misissuance, BGP policy mistakes, and route leaks are unchanged.

A rollback to the Current Suite is a recovery mechanism only while that suite remains trustworthy and policy permits its use. After the Current Suite becomes forgeable, or after a deployment adopts a policy that permits only the Next Suite, such a rollback is a downgrade rather than recovery. Where a transition phase permits fallback to Current Suite products, accepting them after a CRQC makes that suite forgeable can admit forged products; Section 11 of [RFC6916] already identifies transition-period downgrade risks.

Divergent algorithm policies and downgrade behavior are primary concerns during a long transition. Divergent suite-selection policies across the RP population can cause different RPs to derive different VRP sets from the same repository. This systemic risk of the transition itself persists for as long as classical and post-quantum algorithm suites coexist.

Parallel Publication can produce different validation results between the Current Suite and Next Suite branches. For example, the RSA branch and the PQC branch might contain different ROA payloads, stale manifests, or different CRL state. The experiment compares validated outputs between the two suites but does not evaluate all possible forms of divergence.

A Mixed Tree rooted in a Current Suite trust anchor is not quantum resistant as a complete certification path. Validation of the certification path depends on every certificate signature along it, including signatures made under the Current Suite. If an adversary can forge a Current Suite certificate signature above the migration boundary, the adversary can substitute a different Next Suite child key and construct a forged subtree.

Mixed Tree migration is therefore suitable only while Current Suite certificate signatures remain trustworthy. It does not provide a recovery procedure after those signatures become forgeable, nor does it by itself establish a quantum-resistant trust anchor. Achieving end-to-end post-quantum security ultimately requires removing dependence on the Current Suite from the complete certification path. Trust-anchor migration and establishment of such a path are separate concerns from the Mixed Tree migration evaluated here.

Larger public keys, signatures, certificates, CRLs, and CMS objects increase the network and processing resources required for repository retrieval and validation. A hostile or misbehaving publication point can impose disproportionate transfer and CPU cost on RPs, and PQC object sizes raise the ceiling of that cost. Resource limits and operational measurements of object size, object count, validation time, and memory use therefore belong in a production readiness evaluation. Production-network measurements of RRDP, rsync, and Erik Synchronization under projected repository sizes and churn rates are needed before large-scale deployment.

HSM implementations of PQC algorithms are newer than their software counterparts and may lag in side-channel hardening, fault-attack resistance, and certification. A CA key that is protected against extraction but is used by an implementation that leaks sensitive information through side channels does not receive the intended level of protection. Side-channel resistance is algorithm- and implementation-dependent and requires separate evaluation for each candidate and platform.

ML-DSA supports both deterministic and randomized signing. Signatures generated using either mode are interoperable. The experiment uses the signing mode selected by the implementation and does not compare the two modes. Randomized signing can make implementations easier to harden against fault and hardware side-channel attacks [RFC9881].

Algorithm confusion is possible if AlgorithmIdentifier parameters, SignerInfo digestAlgorithm, CMS signed attributes, or certificate SubjectPublicKeyInfo encodings are inconsistently handled. The experimental validation criteria reject malformed AlgorithmIdentifier encodings and follow the parameter rules of the referenced LAMPS specifications.

The LAMPS composite construction provides a conditional EUF-CMA guarantee only when every component is verified, at least one component remains EUF-CMA secure, and the prehash remains collision resistant. It does not generally preserve SUF-CMA from only one strongly unforgeable component. The experiment follows the component-key separation rules in [I-D.ietf-lamps-pq-composite-sigs]; reuse as standalone keys or in other composite combinations can enable stripping and cross-protocol attacks. A defect in one component is tolerated only if the other component and the combiner, parser, encoding, key management, and verification of all component signatures are unaffected. A shared implementation defect or compromise of both component keys is not mitigated by the composite construction. After a CRQC breaks ECDSA, the composite suite's unforgeability depends on ML-DSA-65.

15. IANA Considerations

This document requests no IANA actions. It reuses algorithm identifiers defined by the referenced LAMPS specifications and defines no new RPKI object type, file extension, or content type.

16. Questions for Further Work

The following questions require further discussion and implementation evidence.

- Which post-quantum signature suite offers an appropriate balance of security, object size, validation performance, implementation maturity, and HSM support for the RPKI?
- How should the algorithms used for EE certificates and CMS signatures be related, and should alternative constructions such as the Null Scheme be considered?
- Under what conditions should Parallel Publication or Mixed Tree migration be used, and how should RP readiness and migration timing be determined?
- How should provisioning protocols support a Next Suite child CA under a Current Suite parent, including certificate requests, proof of possession, and the algorithm-suite associations defined by RFC 6916?
- How should RPKI trust anchors and the separate BPKI used for provisioning and publication be migrated while maintaining trust and operational continuity?
- Can RRDP and rsync accommodate the increased repository size and publication activity, or is a more selective synchronization protocol such as Erik needed?

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Appendix A. Measurement Details

A.1. Experimental Artifacts and Environment

The designated evidence snapshot is the GitHub release `draft-yoshikawa-sidrops-pqc-rpki-03` [pqc-rpki-lab]. It includes code and results for the cryptographic, RP, repository, and RRDP experiments, with source manifests recording origins, observation times, and hashes. Large raw datasets are kept outside Git; exact snapshot reanalysis requires the retained bodies if the source URLs have expired.

Cryptographic benchmarks used an Apple M4 running macOS 26.5.2, OpenSSL 3.6.2, and liboqs 0.15.0. The primary algorithm benchmark was compiled with optimization level -O2, while the Composite ML-DSA benchmark was compiled with -O3. The latter implements revision 19 of [I-D.ietf-lamps-pq-composite-sigs].

The X.509, CMS, and RP experiments used OpenSSL 3.6.2 and the experimental Composite implementation recorded in the evidence snapshot. End-to-end validation was also exercised on a 12-vCPU x86-64 host using rpki-client 9.8. The RSA, pure ML-DSA-65, Composite, and Mixed Tree repositories all produced the expected VRPs.

A.2. Repeated Cryptographic Operation Timing

The table reports the median and sample standard deviation across ten repetitions of 1,000 operations on a fixed 32-byte message. Values are wall-clock seconds per 1,000 operations.

Algorithm	Sign median	Sign stdev	Verify median	Verify stdev
RSA-2048/SHA-256	0.340326	0.002114	0.009792	0.000057
P-256/SHA-256	0.012660	0.000332	0.034535	0.005361
Ed25519	0.016657	0.006613	0.040442	0.018694
ML-DSA-44	0.249238	0.029970	0.047798	0.002460
ML-DSA-65	0.405729	0.017570	0.073411	0.014139
ML-DSA-87	0.477243	0.012468	0.115963	0.001536

Table 2

The measurements show that RSA-2048 is substantially faster at verification than the other evaluated algorithms in this environment. ML-DSA verification cost increases with the parameter set, with ML-DSA-65 requiring about 7.5 times the RSA verification time and ML-DSA-87 about 11.8 times. Signing shows a different pattern: ML-DSA-44 is faster than RSA-2048, while ML-DSA-65 and ML-DSA-87 are somewhat slower. The Ed25519 verification samples show substantial variability, as reflected in the reported standard deviation.

These are primitive-operation timings, not complete RP-validation timings.

The benchmark also covers 512-byte, 2-KiB, and 8-KiB messages and records key-generation timing, variance, and process peak RSS. Those raw results are retained in [pqc-rpki-lab].

A.3. Composite ML-DSA Operation Measurements

Each configuration was measured in one run of 100,000 signing and 100,000 verification operations. The timings include message-representative construction, ML-DSA context binding, both component signature operations, raw key and signature concatenation, and verification of all component signatures. The ML-DSA-87 configuration uses P-384 because revision 19 of [I-D.ietf-lamps-pq-composite-sigs] does not define ML-DSA-87 with P-256.

Composite	Sign (s/100k)	Verify (s/100k)	PubKey (B)	Mean sig (B)
ML-DSA-44 + P-256	26.0	8.3	1377	2491
ML-DSA-65 + P-256	45.6	11.9	2017	3380
ML-DSA-87 + P-384	59.4	32.8	2689	4730

Table 3

The measurements show that both cryptographic cost and key and signature sizes increase with the ML-DSA parameter set. For the evaluated ML-DSA-65 + P-256 configuration, the classical component adds relatively little to key and signature sizes compared with ML-DSA-65 alone, while verification must still perform both component checks.

The timings exclude key generation, file I/O, X.509, CMS, RP processing, and HSM latency.

A.4. Measured Complete Object Sizes

The laboratory fixtures in [pqc-rpki-lab] were generated with OpenSSL 3.6.2 and the existing Composite provider. They contain complete DER-encoded RFC 6487-profiled certificates with [RFC3779] resource extensions, CRLs, ROA CMS objects, and manifest CMS objects. CMS sizes include the embedded EE certificate, signed attributes, and signature.

Suite	CA cert (B)	CRL (B)	ROA CMS (B)	Manifest CMS (B)
RSA-2048	1064	415	1621	1743
ML-DSA-44	4264	2575	7016	7138
ML-DSA-65	5793	3464	9434	9556
ML-DSA-65 + P-256 Composite	5927	3533	9637	9758

Table 4

The laboratory manifests contain two fixed fileList entries; the repository projection instead preserves observed fileLists. The Composite sizes reflect variable-length ECDSA signatures in this capture; the projection uses the maximum signature size stated below.

The earlier, separate certificate/CRL measurements for classical reference algorithms, ML-DSA-87, SLH-DSA, and experimental Falcon-512 are retained in [pqc-rpki-lab]. The Falcon-512 measurements use oqs-provider 0.11.0-rc1 with liboqs 0.15.0 and an experimental OID and encoding, not a final FN-DSA PKIX profile. Those supplementary measurements are not inputs to the repository and RRDP projections.

A.5. Source-Calibrated Repository Projection

The Global RPKI Report for Q3 2026 [Global-RPKI-Q3-2026] reports initial-state objects (Tables 1-2), newly discovered objects (Tables 5-10), and RRDP deltas (Tables 3-4) as distinct populations. Object counts differ from VRP counts, and discovery does not establish issuance time. Rounded values and unspecified sampling instants prevent exact reconstruction of its merged multi-vantage baseline; discrepancies are documented in [pqc-rpki-lab].

The byte baseline is the RPKIViews dango cache archive labeled 2026-09-30T23:49:58Z [RPKIViews-Cache-20260930]. It contains 565,166 objects totaling 1,092,318,658 DER bytes; an independent tar-header scan reproduced both totals. This is one collector's cache, not the

complete merged global RPKI or a new independent RPKI validation. Objects are counted by archived path, including five trust-anchor cache certificates; embedded EE certificates are counted only within CMS objects.

Object type	Count	DER MiB
ASA (ASPA)	3,352	5.98
CER	55,089	82.89
CRL	54,963	44.18
MFT	54,963	140.27
ROA	396,799	768.41

Table 5

The laboratory fixtures passed certificate-chain, CRL-signature, and CMS checks, but independent cryptographic interoperability remains unverified. The projection re-encodes each observed object with the candidate key, signature, AlgorithmIdentifier, and CMS digest-attribute lengths, accounting for DER length boundaries. It preserves payloads, names, resource extensions, revoked entries, and manifest fileLists and hash lengths. Observed manifests contain 1 to 48,932 entries and CRLs up to 65,312 revoked entries. BGPsec subject public keys remain unchanged. The evaluated Composite projection uses a conservative maximum 3,381-byte encoded signature because its ECDSA component is variable length. Projected key, signature, and digest values are placeholders for size calculation; these are not valid re-signed PQ repositories.

The table compares the total DER footprint of the observed repository under three scenarios: full replacement with the Next Suite, idealized parallel publication of both suites, and migration of the largest eligible CA subtree while the rest remains on RSA.

Suite	Full replacement GiB	Idealized parallel GiB	Mixed subtree GiB
RSA-2048	1.017	2.035	1.017
ML-DSA-44	3.579	4.596	2.302
ML-DSA-65	4.728	5.745	2.878
ML-DSA-65 + P-256 Composite	4.825	5.842	2.927

Table 6

The RSA parallel row is a duplicate-baseline control. This is a hypothetical steady-state footprint, not an RFC 6916 phase schedule or operational deployment procedure. The Mixed Tree scenario selects the largest observed non-self-issued CA subtree with a known parent, using SKI/AKI links:

9,873 CA keys and 265,585 affected objects. The boundary certificate keeps its RSA parent signature while its subject key changes; descendant signatures change only where their issuer is in the selected subtree. These links describe a structural hierarchy; they do not establish resource/path validity or migration readiness. The selected boundary and exact byte totals are recorded in [pqc-rpki-lab].

A.6. Repository Transport Measurements

The assessment uses observed publication activity; it does not assume ROA churn or apply a multiplier to aggregate RRDP traffic. The September delta sample comes from the RPKIViews raw RRDP archive [RPKIViews-RRDP]. It contains 6,953 documents across 13 notification hosts, covering complete serial intervals after the first and through the last observations in the 2026-09-30 collector log. The collector log's timezone is unspecified. The sample contains 190,738 publish elements: 93,357 manifests, 93,265 CRLs, 3,499 ROAs, 545 certificates, and 72 ASPA objects. Of these publishes, 189,289 contain a replacement hash; the sample also contains 768 withdrawal elements. Repeated publishes are counted, including periodic republication and replacement.

A separate live snapshot sample was captured on 2026-10-27 between 00:16 and 00:19 UTC from all 39 distinct notification URLs at the selected September hosts. This October cohort is separate from the September data and does not represent a simultaneous global state. All bodies were uncompressed and matched their notifications' SHA-256, session, and serial. The 39 snapshot documents contain 569,070 publish elements and total 1,565,796,047 XML bytes. A 496-byte text TAL published in the AFRINIC snapshot is included in transfer bytes, excluded from DER totals, and held at its observed size; future TAL contents are outside this model.

The table compares RRDP XML response-body sizes across signature suites: the sum of 6,953 captured delta documents and the sum of one snapshot from each of 39 endpoints. The dates identify the datasets; neither column represents daily per-client traffic.

Suite	Sep 30 captured deltas (GiB)	Oct 27 captured snapshots (GiB)
RSA-2048	1.834	1.458
ML-DSA-44	2.736	4.888
ML-DSA-65	3.131	6.426
ML-DSA-65 + P-256 Composite	3.164	6.556

Table 7

RSA values are captured XML body sizes; PQ values are estimates from the same observed object contents. For each publish element, the model changes its base64 length by 4*ceil(new_DER_bytes/3) - 4*ceil(old_DER_bytes/3), retaining the original URI, XML framing,

whitespace, and withdrawal elements. RSA identity projections reproduce the captured byte totals exactly. Snapshot payloads were parsed, without full schema or RPKI resource/path validation.

The snapshot column counts one copy of each selected body for bootstrap or fallback, not recurring traffic. The columns cover different dates and must not be added or used to infer a snapshot-to-delta traffic ratio. The estimates exclude notifications, compression, HTTP/TLS overhead, retries, polling, client catch-up, and snapshot fallback frequency. Neither cohort covers a full quarter or all publication points; no quarterly or all-client traffic total is inferred. Equivalent trace-backed rsync and Erik comparisons remain separate work. Collection and validation details, exact totals, and retention limits are recorded in [pqc-rpki-lab].

A.7. Open Measurement Tasks

Remaining measurement tasks include:

- CA key rollover, publication cycle, and full validation across the evaluated configurations over a multi-CA corpus modeled on the public RPKI that can be re-signed under each evaluated configuration. The controlled single-child and synthetic 100-child results do not reproduce public RPKI topology.
- Full-quarter and all-publication-point RRDP transfer, snapshot fallback frequency, compression, polling-interval, and cache behavior, and equivalent trace-backed rsync and Erik comparisons. The captured XML bodies and length projections do not measure per-client WAN traffic or throughput.
- Long-running validator memory and cache growth. The current results record process peak RSS for bounded executions only.
- HSM performance and support.

Appendix B. Changes from -02

This section is to be removed before publication as an RFC.

- Clarified the experimental scope and expanded the discussion of premature algorithm selection, migration costs, and operational risks.
- Updated the implementation status of cryptographic libraries and clarified the limitations of experimental Krill issuance, CSR/PoP processing, and RP interoperability.
- Replaced the synthetic repository-size model with complete-DER size projections based on an observed RPKI cache and real object structures.
- Added repository-footprint comparisons for idealized Parallel Publication and Mixed Tree migration using the observed CA hierarchy.
- Replaced synthetic transport evaluations with projections based on captured RRDP delta activity and snapshot data. Equivalent rsync and Erik Synchronization comparisons remain future work.
- Clarified the distinction between measured and projected results, including data coverage, assumptions, limitations, and reproducibility.

- Simplified the presentation of earlier experiments while retaining relevant RP validation results and updating the supporting experimental evidence.

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